

Prabesh Bista

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🌐 <https://prabeshbista.github.io/>

Education

- 03/2026 – Present 📖 **Ph.D. in Physics**
Stony Brook University, New York, USA
- 09/2021 – 11/2024 📖 **M.Sc. in Physics**
National Central University, Zhongli, Taiwan
Thesis: *Development of a Non-Collinear Optical Parametric Amplifier for Photoelectron Spectroscopy*.
Advisor: Dr. Cheng-Tien Chiang
- 09/2015 – 09/2019 📖 **B.Sc. in Physics**
Tribhuvan University, Kathmandu, Nepal

Research Publications

Journal Articles

- 1 M. Singh, **P. Bista**, Y.-C. Lin, C.-N. Kuo, Y.-J. Chuang, M. Paleschke, C.-B. Huang, M.-C. Chung, C.-S. Lue, and C.-T. Chiang, “Valence band momentum imaging of NiTe₂ by two-photon photoemission momentum microscopy”, Applied Physics Letters **128**, 031601 (2026).
- 2 T.-I. Yang, Y.-W. Huang, **P. Bista**, C.-F. Ding, J. Chen, C.-T. Chiang, and H.-C. Chang, “Photoluminescence of nitrogen-vacancy centers by ultraviolet one-and two-photon excitation of fluorescent nanodiamonds”, The Journal of Physical Chemistry Letters **13**, 11280–11287 (2022).

Presentation

Poster

- 09/2023 📖 Thematic summer school on 2D Materials in Roscoff, France
Title: *Non-collinear optical parametric amplifier for photoemission spectroscopy*
- 01/2023 📖 Annual Meeting of the Physical Society of Taiwan
Title: *Development of an Infrared non-collinear optical parametric amplifier at sub-megahertz repetition rates*
- 11/2022 📖 IAMS Young Fellow Research Presentation
Title: *Construction of an infrared by non-collinear optical parametric amplifier*

Experience

Research Experience

- 04/2026 – Present 📖 **Graduate Research Assistant**
Allison Group, Stony Brook University
• Currently working on building and aligning an optical setup for time-resolved photoemission spectroscopy.
- 10/2025 – 03/2026 📖 **Intern**
Sunko Group, Institute of Science and Technology Austria
• Characterized shot noise in a balanced photodetector system.
• Aligned the laser beam path into a cryostat.

Experience (continued)

- 12/2024 – 05/2025  **Intern**
Baykusheva Group, Institute of Science and Technology Austria
- Built a non-degenerate parametric down-conversion (PDC) setup using a non-linear crystal.
 - Developed a LabVIEW-based control and data-acquisition program for an Ophir power meter.
 - Characterized beam-pointing stability.
- 09/2021 – 11/2024  **Master's Research Project**
Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan
- Constructed a non-collinear optical parametric amplifier from the ground up.
 - Investigated the electronic structure of bulk NiTe₂ using ARPES.
- 08/2020 – 08/2021  **Research Assistant**
Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan
- Generated second- and third-harmonic signals in nonlinear optical experiments.
 - Developed a LabVIEW program for a Thorlabs PM100D power meter.
 - Supported laboratory setup and component procurement.
- 03/2020 – 07/2020  **Intern**
Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan
- Used Kelvin probe and ambient photoemission spectroscopy to determine the valence-band maximum of Pt- and Ti-doped WS₂.

Mentorship

IAMS-IIP Internship

- 06 – 07/2023  **Marucheth Thongtheppairoj**
Undergrad student at Sirindhorn international institute of Technology (SIIT), Thailand

Honors and Awards

- 09/2021–08/2024  **TIGP Scholarship** (Academia Sinica, Taiwan) — full scholarship covering tuition and living expenses.
- 11/2018  **Outstanding Administrative Staff Award** — Student Welfare Society, Sagar-matha Engineering College.
- 09/2018  **Travel Grant** — National Center for Physics, Pakistan; supported participation in the workshop on Tracking Detector in High Energy Physics.

Skills

- Programming  Python, LabVIEW
- Software  Microsoft Office, L^AT_EX, Origin, SOLIDWORKS, ImageJ
- Optics  Beam profiling, second- and third-harmonic generation, white-light continuum generation, optical parametric amplification, pulse-width compression and characterization
- Other  Hands-on experience with angle-resolved photoemission spectroscopy (ARPES)

Activities and Outreach

-  Attended the workshop on [Transport and Optics in Topological Systems](#) at the Institute of Physics, Academia Sinica, Taipei, Taiwan (June 15–17, 2024).
-  Participated in the summer school on [Advanced Physics of van der Waals Hetero-Structures](#) in Roscoff, France (September 23–October 1, 2023).
-  Volunteered in an [intercultural service](#) program, supporting a school in Taiwan through online engagement (September 5–December 1, 2022).

Activities and Outreach (continued)

- Participated in the online [FOCUS PEEM](#) workshop organized by FOCUS GmbH (June 16–17, 2021).
- Attended the Workshop on Space Weather and Upper Atmospheric Physics in Kathmandu, Nepal (September 23–27, 2019).
- Participated in the workshop on [Tracking Detector in High Energy Physics](#) in Islamabad, Pakistan (October 15–19, 2018).
- Organized the National Science Exhibition at Patan Multiple Campus, Lalitpur, Nepal (February 18–19, 2018).